

Guhankarthik A G

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Coimbatore, TN

Final-year EEE student with hands-on expertise in LLM fine-tuning, multi-agent orchestration, and on-device inference. Built and deployed agentic ML pipelines across computer vision, NLP, and time-series domains. Focused on end-to-end ML systems — from transformer internals to containerised deployment.

Technical Skills

- **Languages:** Python, C++
- **AI / ML:** Machine Learning, Deep Learning, Transformers & LLMs, Fine-Tuning, RAG, Agentic Systems (ReAct, Multi-Agent), Quantization, On-Device Inference
- **Frameworks:** PyTorch, HuggingFace, llama.cpp, Weights & Biases, scikit-learn, TensorFlow, FastAPI
- **Databases:** PostgreSQL, FAISS, Qdrant
- **Infrastructure:** Docker, Linux, Git

Experience

Forge Innovation and Ventures

Jan 2026 – Jun 2026

Innovation Graduate Trainee

- Led a team of 6 as Project & Tech Lead to develop an anomaly discovery pipeline using LangGraph, YOLO, VLM-based annotation, FAISS filtering, and HDBSCAN clustering, processing **137 industrial inspection images** into **6 high-cohesion novel defect clusters**.
- Designed a cache-aware closed-loop retraining workflow with dynamic VLM prompting, reducing manual review units by **94% (103 novel crops → 6 representative clusters)** while accelerating iterative experimentation through cached VLM inference.

The Fusion Apps

Jan 2025 – Sep 2025

Machine Learning Engineer Intern

- Evaluated facial analysis pipelines (detection, recognition, enhancement, quality assessment); proposed a landmark-based facial quality-check layer to intercept low-quality frames before recognition, reducing duplicate identities.
- Trained CV models for violence detection (MultiViT-v2 + BiGRU); explored YOLO, Partial FC, and Sub-Center ArcFace architectures.

Projects

AutoAgenticML — Agent-Driven ML Automation Framework

LangGraph · FastAPI · Docker · scikit-learn

- Designed and developed a **hierarchical multi-agent AutoML system** (L0/L1/L2) orchestrating the end-to-end ML lifecycle—data ingestion, profiling, preprocessing, model selection, training, evaluation, and inference—with human-in-the-loop decision checkpoints and deployment via Docker Hub and Render.
- Validated the framework across benchmark classification and regression datasets, autonomously training end-to-end pipelines with minimal user intervention, achieving 77.27% classification accuracy and competitive regression performance without manual hyperparameter tuning.

TinyLLM — Autoregressive Transformer Training from Scratch

PyTorch · Weights & Biases

- Built and trained a 2.75M-parameter decoder-only Transformer from scratch in PyTorch, implementing RoPE positional embeddings, SwiGLU feed-forward layers, and SentencePiece tokenization, achieving 7.55 validation perplexity with 83.2 tokens/sec CUDA inference throughput.

QSystem — Leakage-Aware Time-Series Forecasting & Backtesting

HuggingFace · PyTorch · hmmlearn · CatBoost

- Built a hybrid ML quantitative trading system integrating HMM-based regime detection, NLP sentiment scoring, and CatBoost-based forecasting; validated using rolling walk-forward, fee-aware (0.1% transaction cost) backtesting after identifying and correcting multiple sources of temporal data leakage.
- Backtested across 4,711 historical market intervals, executing 18 simulated trades with a 70% buy-side win rate, 1.64 annualized Sharpe ratio, and 0.06% maximum drawdown.

MedComply — Deterministic AI Healthcare Compliance Engine

HuggingFace · NLTK · NLI · FastAPI

- Developed an LLM-powered NABH gap-analysis system automating compliance assessment across 10 NABH accreditation clauses, generating 30+ structured evidence and recommendation artifacts per report via clause-level explainable analysis.
- Derived compliance scores ranging 25–85% across evaluated hospital documents, reducing an estimated 70–90% of first-pass manual audit effort.

Education

Kumaraguru College of Technology

B.E. Electrical and Electronics Engineering

Certifications

AWS Academy — Machine Learning Foundations | Programming for Everybody — Coursera | Python Data Structures — Coursera